

Theoretical Analysis of Safety Guarantees with Shielding

1 Overview

The current research line studies a *layered shielding framework* for multi-UAV cooperative search. The key distinction is no longer whether safety is turned on or off, but rather which level of conservativeness is active:

- **Hard-safety layer:** one-step safety is always enforced before execution.
- **Recursive-feasibility layer:** the shield prefers actions that preserve at least one future safe action.
- **Look-ahead layer:** the shield can be extended to short-horizon viability checks.
- **Scheduling layer:** training progress and online risk determine how strongly the stricter layers should intervene.

Under this interpretation, the main theoretical question is not whether progressive shielding occasionally permits unsafe actions, but how hard safety, recursive feasibility, short-horizon viability, and risk-aware conservativeness can be combined into a unified and computationally tractable decision process.

2 Safe Set and Shield Operator

Definition 1 (Safe state set). *Let s_t denote the system state at time t , including UAV positions, threat positions, and other necessary environmental variables. The safe state set is defined as*

$$\mathcal{S}_{\text{safe}} = \{s_t \mid \|u_{i,t} - u_{j,t}\| \geq R_U^{\min}, \forall i \neq j; \|u_{i,t} - e_{k,t}\| \geq R_E^{\min}, \forall i, k\}, \quad (1)$$

where $R_U^{\min}$ is the minimum safety distance between any two UAVs, and $R_E^{\min}$ is the minimum safety distance between a UAV and a threat.

Definition 2 (Safe joint action set). *Let A denote the joint action space and let $f(s_t, a_t)$ denote the state transition function. The safe joint action set at state s_t is defined as*

$$A^{\text{safe}}(s_t) = \{a_t \in A \mid f(s_t, a_t) \in \mathcal{S}_{\text{safe}}\}. \quad (2)$$

In other words, a joint action is safe if the next state remains inside the safe state set after applying that action.

Definition 3 (Shield operator). *Define the shield operator as an action-set filter rather than a hard-coded external action selector. Given the current state s_t , the shield produces an allowed action set*

$$\mathcal{A}_t^{\text{allow}}(s_t) \subseteq A^{\text{safe}}(s_t). \quad (3)$$

The actor then re-selects an action from this set through a masked policy

$$\pi_{\theta}^{\text{mask}}(a \mid o_t, \mathcal{A}_t^{\text{allow}}) = \frac{\pi_{\theta}(a \mid o_t) \mathbf{1}[a \in \mathcal{A}_t^{\text{allow}}]}{\sum_{a' \in A} \pi_{\theta}(a' \mid o_t) \mathbf{1}[a' \in \mathcal{A}_t^{\text{allow}}]}, \quad (4)$$

and the final executed action is sampled or selected from π_θ^{mask} . This formulation preserves actor-side preference learning while ensuring that only allowed actions can be executed.

Definition 4 (One-step recursive-feasible action set). *The one-step recursive-feasible action set is defined as*

$$A^{\text{rec}}(s_t) = \left\{ a_t \in A^{\text{safe}}(s_t) \mid A^{\text{safe}}(f(s_t, a_t)) \neq \emptyset \right\}. \quad (5)$$

An action is recursive-feasible if it is safe now and does not immediately lead to a state with no remaining safe action.

Definition 5 (Small-horizon viable action set). *For a finite horizon $H \geq 1$, define the small-horizon viable action set*

$$A_H^{\text{viable}}(s_t) = \{a_t \in A \mid \exists \text{ a safe continuation of length } H \text{ after } a_t\}. \quad (6)$$

This set generalizes one-step recursive feasibility to short-horizon look-ahead safety. In practice, the current implementation may use a greedy or approximate future-safety existence test as a tractable surrogate.

3 One-Step Safety Guarantee

Proposition 1 (One-step safety). *Suppose the current state satisfies $s_t \in \mathcal{S}_{\text{safe}}$, and the allowed action set returned by the shield satisfies*

$$\mathcal{A}_t^{\text{allow}}(s_t) \subseteq A^{\text{safe}}(s_t), \quad (7)$$

with $\mathcal{A}_t^{\text{allow}}(s_t) \neq \emptyset$. If the final executed action is re-selected by the actor from $\mathcal{A}_t^{\text{allow}}(s_t)$, then the next state satisfies

$$s_{t+1} = f(s_t, \tilde{a}_t) \in \mathcal{S}_{\text{safe}}. \quad (8)$$

Proof. Because $\mathcal{A}_t^{\text{allow}}(s_t) \subseteq A^{\text{safe}}(s_t)$, any action $\tilde{a}_t$ selected from $\mathcal{A}_t^{\text{allow}}(s_t)$ must also belong to $A^{\text{safe}}(s_t)$. By definition of $A^{\text{safe}}(s_t)$, this implies

$$f(s_t, \tilde{a}_t) \in \mathcal{S}_{\text{safe}}. \quad (9)$$

Therefore, $s_{t+1} \in \mathcal{S}_{\text{safe}}$. □

4 Recursive Safety Guarantee

Theorem 1 (Recursive safety). *Assume that:*

- (1) *the initial state is safe, i.e., $s_0 \in \mathcal{S}_{\text{safe}}$;*
- (2) *for any $s_t \in \mathcal{S}_{\text{safe}}$, the shield returns a nonempty allowed set satisfying*

$$\mathcal{A}_t^{\text{allow}}(s_t) \subseteq A^{\text{safe}}(s_t), \quad (10)$$

- (3) *the actor always re-selects the final action from $\mathcal{A}_t^{\text{allow}}(s_t)$;*
- (4) *the shield uses a transition model consistent with the actual environment transition.*

Then, for all $t \geq 0$,

$$s_t \in \mathcal{S}_{\text{safe}}. \quad (11)$$

Proof. We use mathematical induction.

Base case. By Assumption (1), $s_0 \in \mathcal{S}_{\text{safe}}$.

Induction step. Assume that $s_t \in \mathcal{S}_{\text{safe}}$ for some $t \geq 0$. By Assumptions (2) and (3), the final action $\tilde{a}_t$ belongs to $A^{\text{safe}}(s_t)$. By Proposition 1, this implies that

$$s_{t+1} = f(s_t, \tilde{a}_t) \in \mathcal{S}_{\text{safe}}. \quad (12)$$

Hence the claim holds for $t + 1$.

By induction, $s_t \in \mathcal{S}_{\text{safe}}$ for all $t \geq 0$. $\square$

5 Robust Safety under Bounded Prediction Errors

In practical systems, the shield may rely on predicted next-step UAV and threat positions. Therefore, safety guarantees should account for prediction errors.

Theorem 2 (Robust safety under bounded errors). *Suppose the prediction errors satisfy*

$$\|\hat{u}_{i,t+1} - u_{i,t+1}\| \leq \epsilon_U, \quad \|\hat{e}_{k,t+1} - e_{k,t+1}\| \leq \epsilon_E, \quad (13)$$

where $\hat{u}_{i,t+1}$ and $\hat{e}_{k,t+1}$ are predicted UAV and threat positions, while $u_{i,t+1}$ and $e_{k,t+1}$ are the corresponding true positions.

Assume the shield enforces the tightened constraints

$$\|\hat{u}_{i,t+1} - \hat{u}_{j,t+1}\| \geq R_U^{\min} + 2\epsilon_U, \quad (14)$$

$$\|\hat{u}_{i,t+1} - \hat{e}_{k,t+1}\| \geq R_E^{\min} + \epsilon_U + \epsilon_E. \quad (15)$$

Then the true next-step positions satisfy the original safety constraints

$$\|u_{i,t+1} - u_{j,t+1}\| \geq R_U^{\min}, \quad (16)$$

$$\|u_{i,t+1} - e_{k,t+1}\| \geq R_E^{\min}. \quad (17)$$

Proof. By the triangle inequality,

$$\|u_{i,t+1} - u_{j,t+1}\| \geq \|\hat{u}_{i,t+1} - \hat{u}_{j,t+1}\| - \|u_{i,t+1} - \hat{u}_{i,t+1}\| - \|u_{j,t+1} - \hat{u}_{j,t+1}\| \quad (18)$$

$$\geq (R_U^{\min} + 2\epsilon_U) - \epsilon_U - \epsilon_U \quad (19)$$

$$= R_U^{\min}. \quad (20)$$

Similarly,

$$\|u_{i,t+1} - e_{k,t+1}\| \geq \|\hat{u}_{i,t+1} - \hat{e}_{k,t+1}\| - \|u_{i,t+1} - \hat{u}_{i,t+1}\| - \|e_{k,t+1} - \hat{e}_{k,t+1}\| \quad (21)$$

$$\geq (R_E^{\min} + \epsilon_U + \epsilon_E) - \epsilon_U - \epsilon_E \quad (22)$$

$$= R_E^{\min}. \quad (23)$$

Therefore, the original safety constraints still hold under bounded prediction errors. $\square$

6 Progressive Conservativeness during Training

The current research line does not interpret progressive shielding as occasionally allowing unsafe actions to pass through. Instead, one-step hard safety is always enforced, while the *conservativeness level* of the shield is gradually strengthened over training.

A convenient hierarchy is

$$\mathcal{A}^{(0)}(s_t) = A^{\text{safe}}(s_t), \quad \mathcal{A}^{(1)}(s_t) = A^{\text{rec}}(s_t), \quad \mathcal{A}^{(2)}(s_t) = A_H^{\text{viable}}(s_t). \quad (24)$$

A time-varying conservativeness index $\kappa(t)$ selects the active allowed action set:

$$\mathcal{A}_t^{\text{allow}}(s_t) = \mathcal{A}^{(\kappa(t))}(s_t). \quad (25)$$

Thus, the progressive mechanism upgrades the decision process from one-step safety to recursive feasibility and then to short-horizon viability, without relaxing the hard safety constraint itself.

When the actor's proposed action does not belong to the current allowed action set, the shield is triggered and the actor re-selects within $\mathcal{A}_t^{\text{allow}}(s_t)$. A trigger-based penalty can then be added to the reward:

$$r'_t = r_t - \beta_t \mathbb{I}_{\text{shield}}(t). \quad (26)$$

This makes progressive shielding compatible with policy learning: the shield restricts unsafe or overly aggressive decisions, while the actor is still responsible for choosing within the permitted action set.

Remark 1. *Under this interpretation, progressive shielding is a progressive conservativeness mechanism rather than a relaxation of hard safety. Its role is to control how strongly recursive-feasible or look-ahead constraints influence the actor over the course of training.*

7 Risk-Aware Dual Scheduling

The next step is not to directly train a learned risk network, but to upgrade the current binary high-risk gate into a *continuous, interpretable, and low-cost* risk score. The first formal version is

$$\xi_{i,t} = w_1 \xi_{i,t}^{\text{clear}} + w_2 \xi_{i,t}^{\text{region}} + w_3 \xi_{i,t}^{\text{hist}}, \quad w_1 + w_2 + w_3 = 1, \quad w_k \geq 0. \quad (27)$$

A practical initialization is $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$, so that geometric clearance is treated as the dominant term.

The three components are defined as follows.

Clearance risk. Let $m_{i,t}$ denote the current minimum candidate safety clearance (the same quantity used by the existing rule-based gate). Define

$$\xi_{i,t}^{\text{clear}} = \text{clip} \left(1 - \frac{m_{i,t}}{M_c}, 0, 1 \right), \quad (28)$$

where M_c is a normalization constant.

Region risk. This term captures whether the agent is currently located in a locally risky area, such as near the boundary, close to threats, or inside a crowded UAV neighborhood. A simple form is

$$\xi_{i,t}^{\text{region}} = \frac{\mathbb{I}_{\text{boundary}} + \tilde{n}_{i,t}^{\text{threat}} + \mathbb{I}_{\text{crowded}}}{3}, \quad (29)$$

where $\tilde{n}_{i,t}^{\text{threat}}$ is the normalized local threat count.

History risk. Let W be a recent time window. Define the trigger-frequency term

$$\xi_{i,t}^{\text{hist}} = \frac{1}{W} \sum_{\tau=t-W+1}^t \mathbb{I}_{\text{shield}}(i, \tau), \quad (30)$$

which reflects whether the current agent has been repeatedly corrected by the shield in the recent past.

The key motivation for using these three terms first is computational: they are already available or cheaply computable from the current shield pipeline, so they do not require an additional expensive online enumeration of A^{safe} .

At the current stage, the risk score is primarily used to decide whether an agent should remain in one-step hard-safe mode or be upgraded to recursive-feasible checking. A simple scheduling rule is

$$\xi_{i,t} > \eta \quad \Rightarrow \quad \text{activate stronger shield logic for agent } i. \quad (31)$$

This keeps the first risk-aware design tightly coupled to the current implementation.

TODO items for later extensions. Two additional terms are intentionally postponed:

- **Feasibility-proxy risk:**

$$\xi_{i,t}^{\text{feas-proxy}} = 1 - \frac{|A_i^{\text{rule}}(s_t)|}{|A_i|}, \quad (32)$$

or a top- k survival-ratio variant.

- **Uncertainty / preference-conflict risk:** based on whether the actor’s preferred actions are rejected by the rule mask, or on a small action-gap / logit-gap indicator.

These terms are left as explicit TODOs because the first goal is to validate that the low-cost three-term score already provides useful scheduling behavior.

8 Recommended Theoretical Positioning

For the paper, the cleanest theoretical positioning is:

- **Hard safety layer:** one-step safety is always enforced through action-set filtering and actor-constrained re-selection.
- **Conservativeness layer:** recursive-feasible and small-horizon look-ahead sets are used to strengthen future viability when needed.
- **Scheduling layer:** training progress and online risk determine how strongly the stricter shield logic should intervene.

This positioning preserves a clear separation of roles: hard safety provides the basic guarantee, recursive-feasible and look-ahead shielding improve sustainable safety, and progressive/risk-aware scheduling controls how much conservativeness is injected during learning.

9 Summary

The main conclusions are as follows:

- (1) If the final executed action is always re-selected from an allowed set contained in $A^{\text{safe}}(s_t)$, one-step safety can be guaranteed.
- (2) If the initial state is safe and the shield always returns a nonempty allowed set contained in $A^{\text{safe}}(s_t)$, recursive safety follows by induction.
- (3) If prediction errors are bounded, robust safety can be recovered by tightening the shield constraints.

- (4) One-step recursive-feasible shielding and small-horizon look-ahead shielding strengthen *sustainable safety* by avoiding immediate transitions into dead-end states.
- (5) Progressive and risk-aware scheduling should be interpreted as mechanisms for controlling conservativeness, intervention frequency, and action-space shrinkage — not as permissions to violate hard safety.

Therefore, the current research line is best summarized as a layered framework: one-step hard safety is always enforced, recursive-feasible and short-horizon viable sets improve future feasibility, and progressive/risk-aware scheduling determines how aggressively these stronger filters shape learning and execution.

10 Current Research Roadmap

The current plan is to keep the original three-point research line rather than switching the main contribution to learned safe masks. Concretely, the next scientific stages are:

- (1) establish formal baselines for **off**, **safe**, and **recursive**;
- (2) formalize the current rule-based gate as a continuous risk function using the low-cost *clear + region + hist* structure;
- (3) use this risk function to decide when the shield should escalate from one-step hard safety to recursive-feasible checking, and then replace the current greedy future-safe existence approximation with an $H = 2$ small-horizon look-ahead shield;
- (4) introduce progressive conservativeness so that the shield evolves from one-step safety to recursive-feasible and then short-horizon viable filtering over training;
- (5) finally extend the system to full dual scheduling of intervention strength and action-space shrinkage.

An important implementation lesson is that exact recursive shielding should not be naively expanded into full joint future-action enumeration at every step. However, this computational concern refines the *implementation strategy*, not the main scientific direction. The main contribution remains the unified combination of progressive shielding, recursive-feasible / short-horizon viability maintenance, and risk-aware dual scheduling.

11 Comparison with Related Shielding Directions

Table 1 summarizes the positioning of the proposed research direction against the most relevant shielding-related lines of work discussed during the literature scan. The goal is not to claim that shielding, look-ahead, or adaptivity are entirely new, but to clarify that the novelty should be placed on the *unified combination* of training-progress-aware shielding, recursive-feasibility preservation, and risk-aware dual scheduling in the multi-UAV cooperative search setting.

Recommended positioning. The planned work should avoid claiming that shielding, look-ahead shielding, or adaptive shielding are individually novel. Instead, the strongest positioning is that the contribution lies in a *unified framework* that jointly addresses:

- training-progress-aware exploration-safety balancing,
- recursive-feasibility preservation to avoid dead-end states, and
- risk-aware dual scheduling of intervention strength and action-space shrinkage

for multi-UAV cooperative search with moving targets and threats.

Table 1: Comparison between the planned framework and closely related shielding directions

Direction / Representative Work	Main Idea	Look-ahead Shield	Adaptive / Progressive Control	Main Difference from Planned Framework
Classical shielded RL	Use a shield to block unsafe actions before execution	Usually no	Usually fixed	Establishes the basic shielding idea, but does not address training-progress-aware intervention, recursive feasibility preservation, or multi-UAV search coupling.
Dynamic multi-agent shielding (e.g., model-based dynamic shielding for safe and efficient MARL, AAMAS 2023)	Dynamically construct/update shields in multi-agent settings; includes model-based and k -step look-ahead elements	Yes	Dynamic shield construction	This is the closest line. The planned framework should distinguish itself by emphasizing training-progress-aware intervention strength, action-space shrinkage scheduling, and recursive-feasibility-oriented control for multi-UAV target search.
Decentralized / shield decentralization (e.g., NeurIPS 2022)	Decompose or decentralize shielding for multi-agent execution	Not the main focus	Limited	Focuses on decentralized realizability of shields rather than progressive training control or feasibility-aware action-space shrinkage.
POMDP / shield decomposition (e.g., RLC 2024)	Extend shielding ideas to partially observable multi-agent environments	Not the main focus	Limited	Shares the partial-observability setting, but does not center on progressive scheduling or the preservation of future safe action availability.
Adaptive shielding (e.g., ADVICE, 2025)	Adjust shielding behavior based on learned safe/unsafe knowledge or black-box	Partial 7	Yes	Related in spirit to adaptive safety intervention, but not specifically organized around dual scheduling of